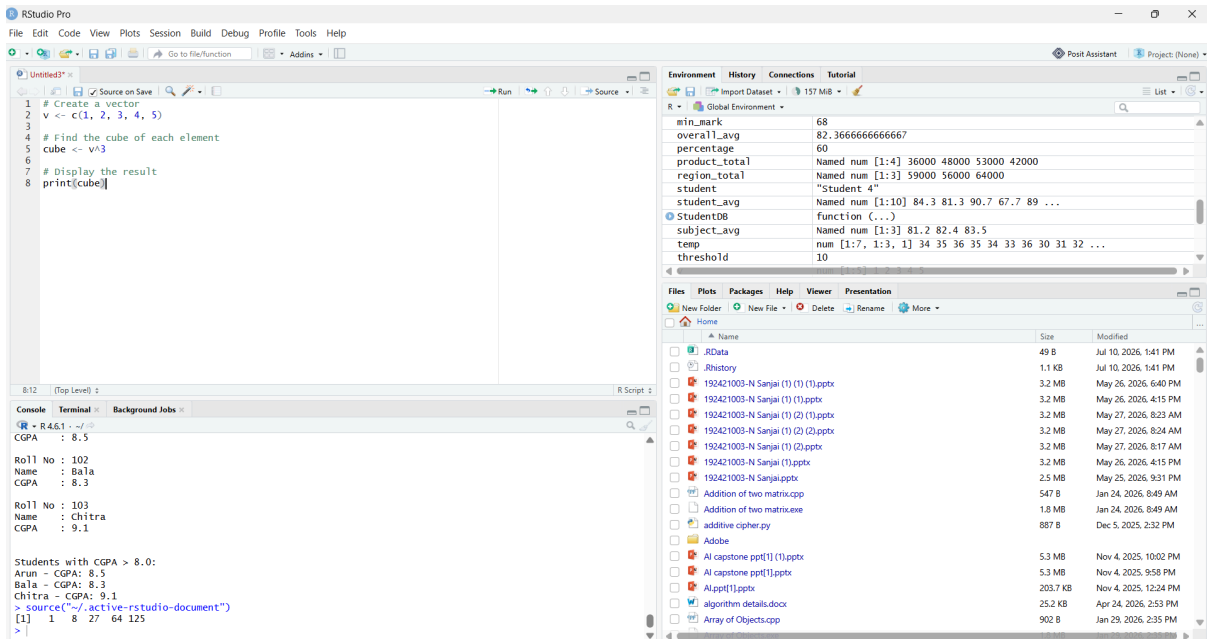


Basic Programs

1)



The screenshot shows the RStudio Pro interface. The script editor on the left contains the following code:

```
1 # Create a vector
2 v <- c(1, 2, 3, 4, 5)
3
4 # Find the cube of each element
5 cube <- v^3
6
7 # Display the result
8 print(cube)
```

The console on the bottom left shows the output of the script:

```
R - R4.6.1 - ~/...
CGPA : 8.5

Roll No : 102
Name : Bala
CGPA : 8.3

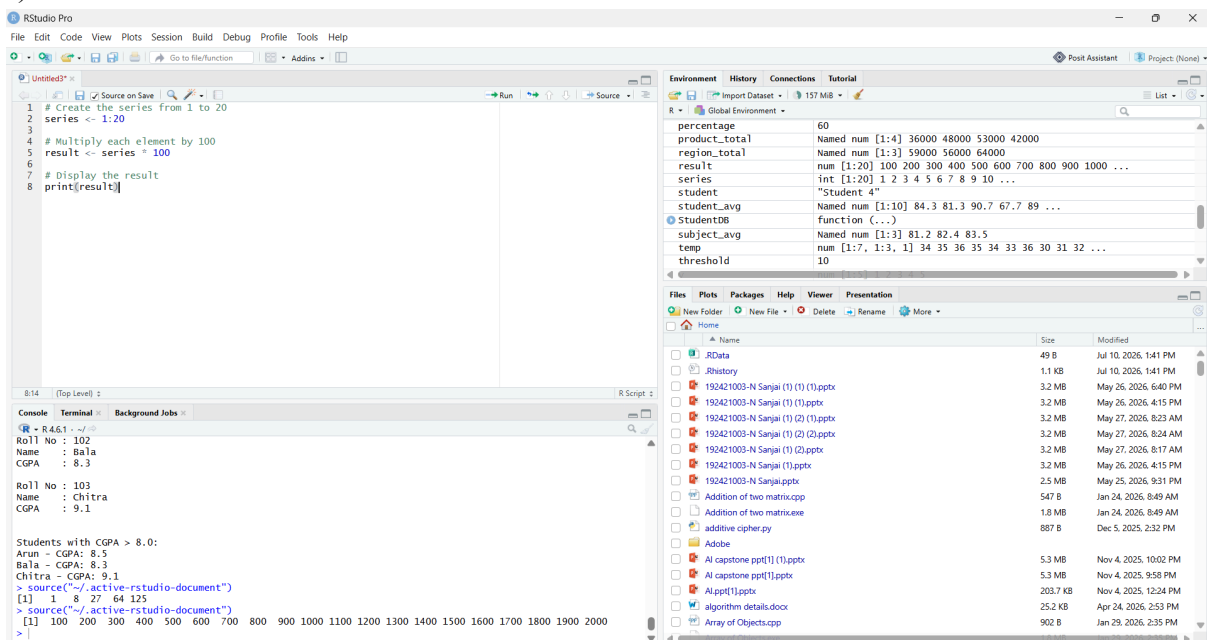
Roll No : 103
Name : Chitra
CGPA : 9.1

Students with CGPA > 8.0:
Arun - CGPA: 8.5
Bala - CGPA: 8.3
Chitra - CGPA: 9.1
> source("~/active-rstudio-document")
[1] 1 8 27 64 125
>
```

The Environment pane on the right shows the following objects:

Object	Value
min_mark	68
overall_avg	82.3666666666667
percentage	60
product_total	Named num [1:4] 36000 48000 53000 42000
region_total	Named num [1:3] 59000 56000 64000
student	"Student 4"
student_avg	Named num [1:10] 84.3 81.3 90.7 67.7 89 ...
StudentDB	function (...)
subject_avg	Named num [1:3] 81.2 82.4 83.5
temp	num [1:7, 1:3, 1] 34 35 36 35 34 33 36 30 31 32 ...
threshold	10

2)



The screenshot shows the RStudio Pro interface. The script editor on the left contains the following code:

```
1 # Create the series from 1 to 20
2 series <- 1:20
3
4 # Multiply each element by 100
5 result <- series * 100
6
7 # Display the result
8 print(result)
```

The console on the bottom left shows the output of the script:

```
R - R4.6.1 - ~/...
Roll No : 102
Name : Bala
CGPA : 8.3

Roll No : 103
Name : Chitra
CGPA : 9.1

Students with CGPA > 8.0:
Arun - CGPA: 8.5
Bala - CGPA: 8.3
Chitra - CGPA: 9.1
> source("~/active-rstudio-document")
[1] 1 8 27 64 125
> source("~/active-rstudio-document")
[1] 100 200 300 400 500 600 700 800 900 1000 1100 1200 1300 1400 1500 1600 1700 1800 1900 2000
>
```

The Environment pane on the right shows the following objects:

Object	Value
percentage	60
product_total	Named num [1:4] 36000 48000 53000 42000
region_total	Named num [1:3] 59000 56000 64000
result	num [1:20] 100 200 300 400 500 600 700 800 900 1000 ...
series	int [1:20] 1 2 3 4 5 6 7 8 9 10 ...
student	"Student 4"
student_avg	Named num [1:10] 84.3 81.3 90.7 67.7 89 ...
StudentDB	function (...)
subject_avg	Named num [1:3] 81.2 82.4 83.5
temp	num [1:7, 1:3, 1] 34 35 36 35 34 33 36 30 31 32 ...
threshold	10

3)

The screenshot shows the RStudio Pro interface. The script in the editor calculates the CGPA for three students (Arun, Bala, Chitra) based on their scores in four subjects. The console output shows the results for each student and a summary of all students.

```

1 # Initialize variables
2 count <- 0
3 num <- 2
4 sum <- 0
5
6 # Add first 10 even numbers
7 while (count < 10) {
8   sum <- sum + num
9   num <- num + 2
10  count <- count + 1
11 }
12
13 # Display the result
14 print(sum)

```

Console Output:

```

R - R4.6.1 ~
CGPA : 8.3

Roll No : 103
Name : Chitra
CGPA : 9.1

Students with CGPA > 8.0:
Arun - CGPA: 8.5
Bala - CGPA: 8.3
Chitra - CGPA: 9.1
> source("~/active-rstudio-document")
[1] 1 8 27 64 125
> source("~/active-rstudio-document")
[1] 100 200 300 400 500 600 700 800 900 1000 1100 1200 1300 1400 1500 1600 1700 1800 1900 2000
> source("~/active-rstudio-document")
[1] 110
>

```

4)

The screenshot shows the RStudio Pro interface. The script in the editor calculates the sum of the first 5 natural numbers. The console output shows the result and a summary of all students.

```

1 # Read the value of n
2 n <- as.integer(readline("Enter n: "))
3
4 # Initialize sum
5 sum <- 0
6
7 # Calculate sum using for loop
8 for(i in 1:n) {
9   sum <- sum + i
10 }
11
12 # Display the result
13 cat("Sum of first", n, "natural numbers =", sum)

```

Console Output:

```

R - R4.6.1 ~
Name : Chitra
CGPA : 9.1

Students with CGPA > 8.0:
Arun - CGPA: 8.5
Bala - CGPA: 8.3
Chitra - CGPA: 9.1
> source("~/active-rstudio-document")
[1] 1 8 27 64 125
> source("~/active-rstudio-document")
[1] 100 200 300 400 500 600 700 800 900 1000 1100 1200 1300 1400 1500 1600 1700 1800 1900 2000
> source("~/active-rstudio-document")
[1] 110
> source("~/active-rstudio-document")
Enter n: 5
Sum of first 5 natural numbers = 15

```

5)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```
1 # Create the vector
2 v <- 11:20
3
4 # Initialize product
5 product <- 1
6
7 # Multiply all elements except the 4th element
8 for(i in 1:length(v)) {
9   if(i == 4)
10     next
11   product <- product * v[i]
12 }
13
14 # Display the result
15 print(product)
```

The console output shows the result of the script:

```
Students with CGPA > 8.0:
Arun - CGPA: 8.5
Bala - CGPA: 8.3
Chitra - CGPA: 9.1
> source("~/active-rstudio-document")
[1] 1 8 27 64 125
> source("~/active-rstudio-document")
[1] 100 200 300 400 500 600 700 800 900 1000 1100 1200 1300 1400 1500 1600 1700 1800 1900 2000
> source("~/active-rstudio-document")
[1] 110
> source("~/active-rstudio-document")
Enter n: 5
Sum of first 5 natural numbers = 15
> source("~/active-rstudio-document")
[1] 47888755200
>
```

The Environment pane shows the following objects:

Object	Class	Attributes
product_total	Named num	[1:4] 36000 48000 53000 42000
region_total	Named num	[1:3] 59000 56000 64000
result	num	[1:20] 100 200 300 400 500 600 700 800 900 1000 ...
series	int	[1:20] 1 2 3 4 5 6 7 8 9 10 ...
student	String	"Student 4"
student_avg	Named num	[1:10] 84.3 81.3 90.7 67.7 89 ...
StudentDB	function	(...)
subject_avg	Named num	[1:3] 81.2 82.4 83.5
sum	num	15
temp	num	[1:7, 1:3, 1] 34 35 36 35 34 33 36 30 31 32 ...
threshold	num	10

6)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```
1 # Initialize variables
2 count <- 0
3 num <- 1
4 sum <- 0
5
6 # Repeat loop
7 repeat {
8   sum <- sum + num
9   num <- num + 2
10   count <- count + 1
11
12   if (count == 10)
13     break
14 }
15
16 # Display the result
17 print(sum)
```

The console output shows the result of the script:

```
Students with CGPA > 8.0:
Arun - CGPA: 8.5
Bala - CGPA: 8.3
Chitra - CGPA: 9.1
> source("~/active-rstudio-document")
[1] 1 8 27 64 125
> source("~/active-rstudio-document")
[1] 100 200 300 400 500 600 700 800 900 1000 1100 1200 1300 1400 1500 1600 1700 1800 1900 2000
> source("~/active-rstudio-document")
[1] 110
> source("~/active-rstudio-document")
Enter n: 5
Sum of first 5 natural numbers = 15
> source("~/active-rstudio-document")
[1] 47888755200
> source("~/active-rstudio-document")
[1] 100
>
```

The Environment pane shows the following objects:

Object	Class	Attributes
product_total	Named num	[1:4] 36000 48000 53000 42000
region_total	Named num	[1:3] 59000 56000 64000
result	num	[1:20] 100 200 300 400 500 600 700 800 900 1000 ...
series	int	[1:20] 1 2 3 4 5 6 7 8 9 10 ...
student	String	"Student 4"
student_avg	Named num	[1:10] 84.3 81.3 90.7 67.7 89 ...
StudentDB	function	(...)
subject_avg	Named num	[1:3] 81.2 82.4 83.5
sum	num	100
temp	num	[1:7, 1:3, 1] 34 35 36 35 34 33 36 30 31 32 ...
threshold	num	10

7)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```
1 # Create a 5x5 matrix
2 mat <- matrix(1:25, nrow = 5, ncol = 5)
3
4 # Display the matrix
5 print(mat)
6
7 # Find the sum of each row using apply()
8 row_sum <- apply(mat, 1, sum)
9
10 # Display the sum of the 3rd row
11 cat("Sum of 3rd row =", row_sum[3])
```

The console output shows the execution results:

```
> source("~/active-rstudio-document")
[1] 110
> source("~/active-rstudio-document")
Enter n: 5
Sum of first 5 natural numbers = 15
> source("~/active-rstudio-document")
[1] 4788755200
> source("~/active-rstudio-document")
[1] 100
> source("~/active-rstudio-document")
      [,2] [,3] [,4] [,5]
[1,]  1   6  11  16  21
[2,]  2   7  12  17  22
[3,]  3   8  13  18  23
[4,]  4   9  14  19  24
[5,]  5  10  15  20  25
Sum of 3rd row = 65
> |
```

The Environment pane shows the following objects:

Object	Class	Value
region_total	Named num [1:3]	59000 56000 64000
result	num [1:20]	100 200 300 400 500 600 700 800 900 1000 ...
row_sum	int [1:5]	55 60 65 70 75
series	int [1:20]	1 2 3 4 5 6 7 8 9 10 ...
student	String	"Student 4"
student_avg	Named num [1:10]	84.3 81.3 90.7 67.7 89 ...
StudentDB	function (...)	
subject_avg	Named num [1:3]	81.2 82.4 83.5
sum	num	100
temp	num [1:7, 1:3, 1]	34 35 36 35 34 33 36 30 31 32 ...
threshold	num	10

8)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```
1 # Create a 5x5 matrix
2 mat <- matrix(1:25, nrow = 5, ncol = 5)
3
4 # Display the matrix
5 print(mat)
6
7 # Find the mean of each column using apply()
8 col_mean <- apply(mat, 2, mean)
9
10 # Display the mean of the 3rd column
11 cat("Mean of 3rd column =", col_mean[3])
```

The console output shows the execution results:

```
> source("~/active-rstudio-document")
[1] 100
> source("~/active-rstudio-document")
      [,2] [,3] [,4] [,5]
[1,]  1   6  11  16  21
[2,]  2   7  12  17  22
[3,]  3   8  13  18  23
[4,]  4   9  14  19  24
[5,]  5  10  15  20  25
Sum of 3rd row = 65
> source("~/active-rstudio-document")
      [,2] [,3] [,4] [,5]
[1,]  1   6  11  16  21
[2,]  2   7  12  17  22
[3,]  3   8  13  18  23
[4,]  4   9  14  19  24
[5,]  5  10  15  20  25
Mean of 3rd column = 13
> |
```

The Environment pane shows the following objects:

Object	Class	Value
mat	int [1:5, 1:5]	1 2 3 4 5 6 7 8 9 10 ...
patients	5 obs. of 4 variables	
pos	int [1, 1:3]	4 3 1
s1	Formal class Student	
s2	Formal class Student	
s3	Formal class Student	
sales	num [1:3, 1:4]	12000 10000 14000 15000 17000 16000 18000 16000 ...
avg_cost	num	29000
balance	num	12000
best_product	String	"Product3"

9)

The RStudio Pro interface displays a script for creating a 5x5 matrix and calculating row medians. The script is as follows:

```
1 # Create a 5x5 matrix
2 mat <- matrix(1:25, nrow = 5, ncol = 5)
3
4 # Display the matrix
5 print(mat)
6
7 # Find the median of each row using apply()
8 row_median <- apply(mat, 1, median)
9
10 # Display the median of the 3rd row
11 cat("Median of 3rd row =", row_median[3])
```

The console output shows the matrix and the median of the 3rd row:

```
Sum of 3rd row = 65
> source("~/active-rstudio-document")
[1,] [,2] [,3] [,4] [,5]
[1,] 1 6 11 16 21
[2,] 2 7 12 17 22
[3,] 3 8 13 18 23
[4,] 4 9 14 19 24
[5,] 5 10 15 20 25
Mean of 3rd column = 13
> source("~/active-rstudio-document")
[1,] [,2] [,3] [,4] [,5]
[1,] 1 6 11 16 21
[2,] 2 7 12 17 22
[3,] 3 8 13 18 23
[4,] 4 9 14 19 24
[5,] 5 10 15 20 25
Median of 3rd row = 13
> |
```

The Environment pane shows the following objects:

Object	Value
max_mark	96
max_temp	41
min_mark	68
n	5L
num	21
overall_avg	82.3666666666667
percentage	60
product	47888755200
product_total	Named num [1:4] 36000 48000 53000 42000
region_total	Named num [1:3] 59000 56000 64000
result	num [1:20] 100 200 300 400 500 600 700 800 900 1000 ...

10)

The RStudio Pro interface displays a script for creating a 5x5 matrix and calculating column sums. The script is as follows:

```
1 # Create a 5x5 matrix
2 mat <- matrix(1:25, nrow = 5, ncol = 5)
3
4 # Display the matrix
5 print(mat)
6
7 # Find the sum of each column using apply()
8 col_sum <- apply(mat, 2, sum)
9
10 # Display the sum of the 5th column
11 cat("Sum of 5th column =", col_sum[5])
```

The console output shows the matrix and the sum of the 5th column:

```
Mean of 3rd column = 13
> source("~/active-rstudio-document")
[1,] [,2] [,3] [,4] [,5]
[1,] 1 6 11 16 21
[2,] 2 7 12 17 22
[3,] 3 8 13 18 23
[4,] 4 9 14 19 24
[5,] 5 10 15 20 25
Median of 3rd row = 13
> source("~/active-rstudio-document")
[1,] [,2] [,3] [,4] [,5]
[1,] 1 6 11 16 21
[2,] 2 7 12 17 22
[3,] 3 8 13 18 23
[4,] 4 9 14 19 24
[5,] 5 10 15 20 25
Sum of 5th column = 115
> |
```

The Environment pane shows the following objects:

Object	Value
mat	int [1:5, 1:5] 1 2 3 4 5 6 7 8 9 10 ...
patients	5 obs. of 4 variables
pos	int [1, 1:3] 4 3 1
s1	Formal class 'Student'
s2	Formal class 'Student'
s3	Formal class 'Student'
sales	num [1:3, 1:4] 12000 10000 14000 15000 17000 16000 18000 16000 ...
avg_cost	29000
balance	12000
best_product	"Product3"

11)

The screenshot shows the RStudio Pro interface. The script editor contains the following R code:

```

1 # Create a 5x5 matrix
2 mat <- matrix(1:25, nrow = 5, ncol = 5)
3
4 # Display the matrix
5 print(mat)
6
7 # Find the standard deviation of each row using apply()
8 row_sd <- apply(mat, 1, sd)
9
10 # Display the standard deviation of the 2nd row
11 cat("Standard Deviation of 2nd row =", row_sd[2])

```

The console output shows the execution results:

```

R - R4.6.1 ~
Median of 3rd row = 13
> source("~/active-rstudio-document")
[1,] 1 6 11 16 21
[2,] 2 7 12 17 22
[3,] 3 8 13 18 23
[4,] 4 9 14 19 24
[5,] 5 10 15 20 25
Sum of 5th column = 115
> source("~/active-rstudio-document")
[1,] [2,] [3,] [4,] [5,]
[1,] 1 6 11 16 21
[2,] 2 7 12 17 22
[3,] 3 8 13 18 23
[4,] 4 9 14 19 24
[5,] 5 10 15 20 25
Standard Deviation of 2nd row = 7.905694
>

```

The Environment pane on the right shows the following objects:

Object	Class	Value
max_temp	num	41
min_mark	num	68
n	num	5L
num	num	21
overall_avg	num	82.3666666666667
percentage	num	60
product	num	47888755200
product_total	Named num [1:4]	36000 48000 53000 42000
region_total	Named num [1:3]	59000 56000 64000
result	num	[1:20] 100 200 300 400 500 600 700 800 900 1000 ...
row_median	int	[1:5] 11 12 13 14 15

12)

The screenshot shows the RStudio Pro interface. The script editor contains the following R code:

```

1 # Create a 5x5 matrix
2 mat <- matrix(1:25, nrow = 5, ncol = 5)
3
4 # Display the matrix
5 print(mat)
6
7 # Convert matrix to a list and apply sqrt()
8 result <- lapply(as.list(mat), sqrt)
9
10 # Display the square root of each element
11 print(result)

```

The console output shows the execution results:

```

R - R4.6.1 ~
[1] 4.472136

[[21]]
[1] 4.582576

[[22]]
[1] 4.690416

[[23]]
[1] 4.795832

[[24]]
[1] 4.898979

[[25]]
[1] 5
>

```

The Environment pane on the right shows the following objects:

Object	Class	Value
result	List of 25	
s1	Formal class Student	
s2	Formal class Student	
s3	Formal class Student	
sales	num	[1:3, 1:4] 12000 10000 14000 15000 17000 16000 18000 16000 ...
avg_cost	num	29000
balance	num	12000
best_product	num	"Product3"
best_region	num	"East"
city_avn	Named num [1:3]	34 7 30 6 39 3

13)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```

1 # Read a number
2 n <- as.integer(readline("Enter a number: "))
3
4 # Find factorial
5 fact <- factorial(n)
6
7 # Display result
8 cat("Factorial of", n, "=", fact)

```

The console shows the output of the script:

```

[[22]]
[1] 4.690416

[[23]]
[1] 4.795832

[[24]]
[1] 4.898979

[[25]]
[1] 5

> source("~/active-rstudio-document")
Enter a number: 6
Factorial of 6 = 720
>

```

The environment pane shows the following objects:

Object	Class	Value
db	<Object containing active bindings>	
eligible	logi [1:5]	FALSE TRUE TRUE FALSE TRUE
eligible_count	3L	
fact	720	
history	chr [1:2]	"Deposited: ₹ 5000" "Withdrawn: ₹ 3000"
i	10L	
ineligible_count	2L	
InventoryValue	num [1:5]	750000 4000 24000 75000 80000
max_mark	96	
max_temp	41	
min_mark	68	

14)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```

1 # Initialize first two Fibonacci numbers
2 a <- 0
3 b <- 1
4
5 # Find the 15th Fibonacci number
6 for(i in 3:15) {
7   c <- a + b
8   a <- b
9   b <- c
10 }
11
12 cat("15th Fibonacci number =", b)

```

The console shows the output of the script:

```

[[22]]
[1] 4.690416

[[23]]
[1] 4.795832

[[24]]
[1] 4.898979

[[25]]
[1] 5

> source("~/active-rstudio-document")
Enter a number: 6
Factorial of 6 = 720
> source("~/active-rstudio-document")
15th Fibonacci number = 377
>

```

The environment pane shows the following objects:

Object	Class	Value
db	<Object containing active bindings>	
eligible	logi [1:5]	FALSE TRUE TRUE FALSE TRUE
eligible_count	3L	
fact	720	
history	chr [1:2]	"Deposited: ₹ 5000" "Withdrawn: ₹ 3000"
i	15L	
ineligible_count	2L	
InventoryValue	num [1:5]	750000 4000 24000 75000 80000
max_mark	96	
max_temp	41	
min_mark	68	

15)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```

1 # First two Fibonacci numbers
2 a <- 0
3 b <- 1
4
5 cat(a, b, " ")
6
7 # Print remaining terms
8 for(i in 3:15) {
9   c <- a + b
10  cat(c, " ")
11  a <- b
12  b <- c
13 }

```

The console output shows the execution results:

```

[[23]]
[1] 4.795832

[[24]]
[1] 4.898979

[[25]]
[1] 5

> source("~/active-rstudio-document")
Enter a number: 6
Factorial of 6 = 720
> source("~/active-rstudio-document")
15th Fibonacci number = 377
> source("~/active-rstudio-document")
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377

```

The Environment pane on the right shows the following objects:

Object	Class	Value
db	<Object containing active bindings>	
eligible	logi [1:5]	FALSE TRUE TRUE FALSE TRUE
eligible_count	3L	
fact	720	
history	chr [1:2]	"Deposited: ₹ 5000" "Withdrawn: ₹ 3000"
i	15L	
ineligible_count	2L	
InventoryValue	num [1:5]	750000 4000 24000 75000 80000
max_mark	96	
max_temp	41	
min_mark	68	

16)

The screenshot shows the RStudio Pro interface. The script editor contains the following code:

```

1 # Create variables
2 a <- 10
3 b <- 20
4 c <- 30
5
6 # Sum using get()
7 total <- get("a") + get("b") + get("c")
8
9 # Display the result
10 cat("Sum =", total)

```

The console output shows the execution results:

```

[[24]]
[1] 4.898979

[[25]]
[1] 5

> source("~/active-rstudio-document")
Enter a number: 6
Factorial of 6 = 720
> source("~/active-rstudio-document")
15th Fibonacci number = 377
> source("~/active-rstudio-document")
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
Sum = 60
> source("~/active-rstudio-document")
Sum = 60
>

```

The Environment pane on the right shows the following objects:

Object	Class	Value
row_median	int [1:5]	11 12 13 14 15
row_sd	num [1:5]	7.91 7.91 7.91 7.91 7.91
row_sum	int [1:5]	55 60 65 70 75
series	int [1:20]	1 2 3 4 5 6 7 8 9 10 ...
student	"student" 4"	
student_avg	Named num [1:10]	84.3 81.3 90.7 67.7 89 ...
StudentDB	function (...)	
subject_avg	Named num [1:3]	81.2 82.4 83.5
sum	100	
temp	num [1:7, 1:3, 1]	34 35 36 35 34 33 36 30 31 32 ...
threshold	10	